

Python – Modular Programming & Functions

MCQs

1. Which keyword is used to define a function in Python?
 - A) function
 - B) define
 - C) def
 - D) fun
2. What is the main purpose of modular programming?
 - A) To make programs longer
 - B) To divide a program into smaller, reusable components
 - C) To avoid using variables
 - D) To eliminate functions
3. Which of the following correctly defines a function?
 - A) function add(a,b):
 - B) def add(a,b):
 - C) define add(a,b):
 - D) func add(a,b):

4. What is the output of the following code?

```
def square(x):  
    return x * x
```

```
print(square(5))
```

- A) 10
 - B) 20
 - C) 25
 - D) 125
5. What is the purpose of the return statement in a function?
 - A) To print the function
 - B) To terminate Python
 - C) To send a value back to the caller
 - D) To define a parameter
 6. What is the output?

```
def add(a, b):  
    return a + b
```

```
x = add(10, 20)  
print(x)
```

- A) 10
 - B) 20
 - C) 30
 - D) None
7. Which of the following is a **parameter**?

```
def multiply(x, y):  
    return x * y
```

- A) multiply
- B) x and y
- C) return

- D) $x * y$
8. In the following statement, what are 10 and 20?

```
add(10, 20)
```

- A) Parameters
 - B) Arguments
 - C) Functions
 - D) Variables only
9. What happens if a function does not contain a return statement?
- A) It always returns 0
 - B) It returns None
 - C) It produces a syntax error
 - D) It returns False
10. What is the output?

```
def test():  
    x = 10
```

```
test()  
print(x)
```

- A) 10
 - B) None
 - C) NameError
 - D) 0
11. The variable x defined inside a function is generally:
- A) A global variable
 - B) A local variable
 - C) A class variable
 - D) A constant
12. Which is an advantage of using functions?
- A) Code reuse
 - B) Better organization
 - C) Easier testing and debugging
 - D) All of the above
13. What is the output?

```
def greet(name="Student"):  
    return "Hello " + name
```

```
print(greet())
```

- A) Hello
 - B) Hello Student
 - C) Student
 - D) Error
14. Which function definition uses a **default parameter**?
- A) `def add(a, b):`
 - B) `def add(a, b=10):`
 - C) `def add(a=10, b):`
 - D) `def add(a, b, c):`
15. What is the output?

```
def calculate(a, b):  
    sum_value = a + b  
    product = a * b  
    return sum_value, product
```

```
x, y = calculate(3, 4)
print(x, y)
```

- A) 7 12
- B) 12 7
- C) 7
- D) Error

16. Which of the following best represents **modular design**?

- A) Writing the entire program inside one large function
- B) Dividing a program into functions such as `read_data()`, `process_data()`, and `display_result()`
- C) Avoiding functions
- D) Using only global variables

17. Consider:

```
def preprocess(data):
    data = data / 10
    return data
```

```
def calculate(data):
    return data * 5
```

```
x = preprocess(20)
y = calculate(x)
print(y)
```

What is the output?

- A) 2
- B) 10
- C) 20
- D) 100

18. Which approach is most appropriate for writing a modular data-processing program?

- A) One large block containing all operations
- B) Separate functions for input, preprocessing, analysis, and output
- C) Only global variables
- D) Repeating the same code multiple times

19. What is the output?

```
def modify(x):
    x = x + 10
```

```
a = 5
modify(a)
print(a)
```

- A) 5
- B) 10
- C) 15
- D) Error

20. Which of the following improves **function reusability**?

- A) Hard-coding values inside the function
- B) Using appropriate parameters
- C) Using many global variables
- D) Duplicating the function

Code-Writing / Practical Questions

21. Write a Python function called `calculate_average()` that accepts a list of numbers and returns their average. Do not use NumPy.
22. Write a Python function called `find_maximum()` that accepts a list of numbers and returns the largest value **without using Python's built-in `max()` function**.
23. Write a modular Python program that contains separate functions to:
 - Read a list of numbers
 - Calculate the mean
 - Calculate the maximum
 - Display the results
24. Write a Python function called `count_even()` that accepts a list of integers and returns the number of even numbers in the list.
25. Write a modular Python program to calculate the **accuracy of a classification model** using two functions:
 - `calculate_accuracy(y_true, y_pred)`
 - `display_result(accuracy)`

Do not use Scikit-learn's built-in accuracy function.

26. Write a Python function that accepts a list of numbers and returns **two values: mean and standard deviation**.
27. Write a function called `normalize()` that accepts a list of numerical values and performs min-max normalization:

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Do not use Scikit-learn's preprocessing functions.

28. Write a modular Python program for a simple **machine-learning preprocessing pipeline** using separate functions:

```
load_data()
↓
remove_missing_values()
↓
normalize_data()
↓
display_data()
```

29. Write a Python function to calculate the **Euclidean distance** between two points without using any built-in distance function.